



भारतीय सूचना प्रौद्योगिकी संस्थान गुवाहाटी
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CS 360: Machine Learning Lab
Practice Assignment 5

Instructions: This is only for practice. Complete it by 12:00 PM today. Your completion will be reviewed by the Teaching Assistants.

1. Download [The Iris species dataset](#) regarding the classification of Iris flower species based on different given predictors. Write a program to do the followings:

- (a) Read the dataset.
- (b) Randomly split the dataset into training and testing splits using k-fold cross validation, for $k=5$.

Hint: Consider 10% of the training set as validation set.

- (c) Design the hypothesis ($\hat{y}$) (both with and without in-built packages) to classify the Iris species using Multi-class Logistic regression to estimation of the model parameters (θ) or (ω) for the different training-testing splits obtained from 1(b). Apply the followings for this:

- Batch gradient descent method for calculating error-gradient.
- One-vs-all strategy for multi-class scenario.

without using in-built Python packages /libraries for both.

- Consider model parameters are to be initialized (θ) or (ω) to zero.
 - Apply hyper-parameter tuning for selecting the best model for the following hyper-parameters, considering the given values:
 1. learning rate, α : $\{0.0001, 0.1\}$
 2. ρ : $\{0.001, 0.01\}$ where, ρ signifies absolute error-difference in two consecutive epochs.
 3. epoch, (T): $\{50, 100\}$
- (d) Calculate and visualize the confusion matrix, considering the designed hypothesis ($\hat{y}$) and the actual targets (y), for all the training: testing splits.
 - (e) Calculate and report the overall accuracy, class-wise accuracy, precision and recall values, with respect to average of the k-folds and the best fold, in separate sheets, and also write the result analysis.